The Major Contributors in the Field of Biology

1. Ernst Mayr :



ERNST MAYR, the Harvard University evolutionary biologist who has been called **'The Darwin of the 20th Century',** was one of the 100 greatest scientist of all time.

He almost single-handedly made the origin of species diversity the central question of evolutionary biology that it is today. **He also pioneered the currently accepted definition of the biological species.**

2. Katherine Esau :



KATHERINE ESAU, reported in her early publications that the curly top virus spreads through a plant via the food-conducting or phloem tissue. **The Anatomy of Seed Plants by Katherine Esau was published in 1960. It was referred to as Webster's of plant biology** — It is encyclopediac.

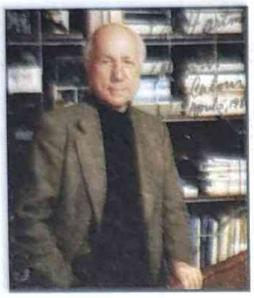
3. G.N. Ramachandran :



G.N. RAMACHANDRAN, an outstanding figure in the field of protein structure, was the founder of the 'Madras School' of conformational analysis of biopolymers. While at Cambridge, Ramachandran met Linus Pauling and was deeply influenced by his publication on models of the α-helix and β-sheet structures that directed his attention to solving the structure of collagen.

His discovery of the triple helical structure of collagen published in Nature in 1954 and his analysis of the allowed conformations of proteins through the use of the 'Ramachandran plot' rank among the most outstanding contribution in structural biology.

4. Melvin Calvin :



MELVIN CALVIN, and co-workers, just after world war II, put radioactivity to beneficial use. **He along with J.A. Bassham studied reaction in green plants forming sugar and other substances from raw materials like carbon dioxide, water and minerals by labelling the carbon dioxide with C^{14}.** Calvin proposed that plants change light energy to chemical energy by transferring an elctron in an organised array of pigment molecules and other substances. **The mapping of the pathways of carbon assimilation in photosynthesis earned him Nobel Prize in 1961.**

5. Alfonso Corti :



ALFONSO CORTI, Italian anatomist, began his scientific career studying the cardiovascular system of reptiles. Later, he turned his attention to the mammalian auditory system. **In 1851, he published a paper describing a structure located on the basilar membrane of the cochlea containing hair cells that convert sound vibrations into nerve impulses, the organ of Corti.**

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1. Panchanan Maheshwari :



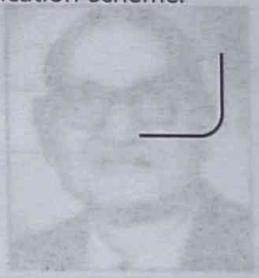
PANCHANAN MAHESHWARI, worked on embryological aspects (plants) and popularised the use of embryological characters (plants) in taxonomy. He established the Department of Botany, University of Delhi as an important centre of research in embryology and tissue culture. His work on test tube fertilisation and intra-ovarian pollination won worldwide acclaim.

He encouraged general education and made a significant contribution to school education by his leadership in bringing out the very first textbooks of Biology for Higher Secondary Schools published by NCERT in 1964.

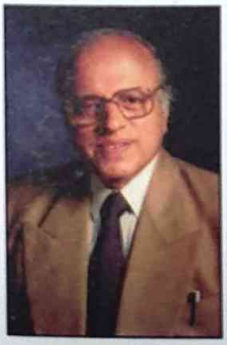
2. James Watson and Francis Crick :



WATSON and CRICK, gave the 3D model of double helical structure for DNA (published in the journal NATURE, 1953) and the replication scheme. **They received the nobel prize in 1962.**



3. M.S. Swaminathan :



M.S. SWAMINATHAN, and his team **at Indian Agricultural Research Institute (IARI) developed the short-duration high-yielding varieties of rice including scented Basmati.** He is also known for the development of the concept of crop cafeteria, crop scheduling and genetically improving the yield and quality.

Swaminathan initiated collaboration with Norman Borlaug, which culminated in the 'Green Revolution' through introduction of Mexican varieties of wheat in India. He is also the initiator of 'Lab-to-Land', food security and several other environmental programmes. **He has been honoured with Padma Bhushan.**

4. Herbert Boyer :



HERBERT BOYER performed studies on a couple of restriction enzymes of the *E.coli* bacterium with especially useful properties. **Boyer observed that these enzymes have the capability of cutting DNA strands in a particular fashion, which left what has became known as 'sticky ends' on the strands.** These clipped ends made pasting together pieces of DNA a precise exercise.

This discovery, in turn, led to a rich and rewarding conversation in Hawaii with a Stanford scientist named **STANLEY COHEN. Cohen had been studying small ringlets of DNA called plasmids** and which float about freely in the cytoplasm of certain bacterial cells and replicate independently from the coding strand of DNA. **Cohen had developed a method of removing these plasmids from the cell and then reinserting them in other cells. Combining this process with that of DNA splicing enabled Boyer and Cohen to recombine segments of DNA in desired configurations and insert the DNA in bacterial cells, which could then act as manufacturing plants for specific proteins. This breakthrough was the basis upon which the discipline of biotechnology was founded.**

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5. Ramdeo Misra :



RAMDEO MISRA is popularly known as **'The Father of Ecology in India'.**

His research laid the foundations for understanding of tropical communities and their succession, environmental responses of plant populations and productivity and nutrient cycling in tropical forest and grassland ecosystems. **Misra formulated the first postgraduate course in ecology in India.**

Due to his efforts, the Government of India established the National Committee for Environmental Planning and Coordination (1972) which, in later years, paved the ways for the establishment of the Ministry of Environment and Forests (1984).